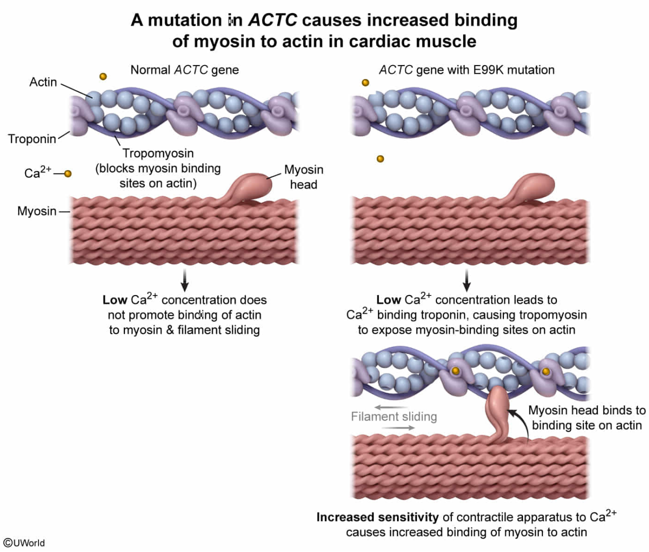
ACTC is a gene coding for cardiac actin. Patients exhibiting a specific mutation in this gene, called the E99K mutation, demonstrate a form of cardiac hypertrophy (cell enlargement) thought to be caused by increased sensitivity of the contractile apparatus to calcium. Given this, patients with the E99K mutation most likely demonstrate increased binding of:

- A. myosin to actin. [50%]
- B. actin to troponin. [10%]
- C. calcium to tropomyosin. [36%]
- D. ATP to calcium. [2%]

Correct
49% answered correctly

Explanation:



Cardiac muscle is responsible for heart wall contractions, which promote the continuous pumping of blood through the cardiovascular system. Cardiac muscle is striated and consists of branched fibers connected to one another by intercalated discs. Cardiac muscle fibers are composed of individual functional units known as sarcomeres. Each sarcomere consists of actin (thin) filaments and myosin (thick) filaments that interact with one another to cause cardiac muscle contraction as follows:

1. At rest, extended heads of the myosin filaments exist in an upright (high-energy) conformation bound to ADP and P_i, whereas actin filaments are bound to two regulatory proteins, troponin and tropomyosin. Troponin is a small protein complex associated with tropomyosin, an elongated protein

wrapped around the actin filament that blocks the myosin binding sites while the sarcomere is at rest.

2. Depolarization of cardiac muscle cells triggers the release of calcium ions from the sarcoplasmic reticulum and from the extracellular fluid into the cytosol. Calcium ions bind troponin, causing a conformational change to tropomyosin described by the sliding filament model.

3. This conformational change exposes the myosin binding sites on the actin filaments, allowing myosin heads to bind the actin filament, forming a cross-bridge. This promotes the release of ADP and P_i from myosin, which in turn changes to the low-energy conformation.

4. These changes cause a power stroke, which acts to drag the actin filament toward the center of the sarcomere.

5. Following the power stroke, a new ATP molecule binds myosin, leading to cross-bridge disassembly.

6. Hydrolysis of this ATP returns the myosin head to its high-energy conformation for the next contraction cycle.

In this question, *ACTC* is a gene encoding a protein comprising cardiac actin filaments. Patients with a specific *ACTC* mutation (E99K) demonstrate a form of cardiac hypertrophy (cell enlargement) thought to be caused by increased sensitivity of the contractile apparatus to calcium. Because of this, a lower intracellular concentration of calcium will be able to bind troponin and cause the conformational change required for the binding of myosin to actin and the subsequent contraction cycle. Therefore, patients with this mutation likely demonstrate **increased binding of myosin to actin**.

(Choice B) Increased sensitivity to calcium would cause decreased, *not* increased, binding of actin to troponin.

(Choice C) Calcium binds troponin, *not* tropomyosin.

(Choice D) ATP binds myosin heads, *not* calcium, following the power stroke during muscle contraction.

Educational objective:
During muscle contraction, calcium binding to troponin facilitates a conformational change in tropomyosin, exposing the myosin binding sites of the

actin filament and allowing sarcomere shortening via the sliding filament model.

Time spent:0 sec
Subject:
Biology

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System:
Musculoskeletal System